Refesh on Mathematics

Introduction to Computer Graphics - Lecture 02

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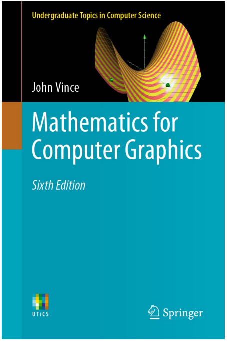
Literature Reference



Mathematics for Computer Graphics
John Vince

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Related Chapters:

- 3 Algebra
- 4 Trigonometry
- 5 Coordinate Systems
- 7 Vectors

Why do we need Mathematics for CG?

Computer Graphics are inherently mathematical.
To render images we need a good foundation of Mathematics.[1]

- Ray-Sphere Intersection $\rightarrow$ Vector-Mathematics
- Projecting a Triangle to screen $\rightarrow$ Matrix-Mathematics
- Coloring and Shading $\rightarrow$ more Vector-Mathematics
- Animation $\rightarrow$ Trigonometry and more Matrices

[1] These are only a view examples for use of Math in CG.

Which Mathematics do we need?

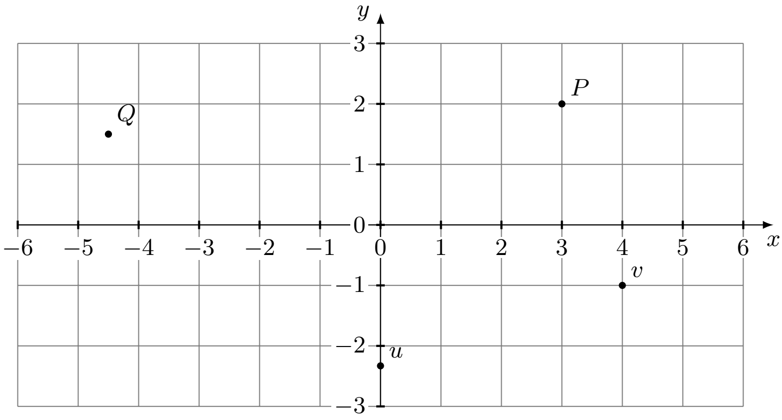
Computer Graphics are inherently mathematical.
To render images we need a good foundation of Mathematics.

Which Mathematics do we need?

Relevant topics include but are not limited to:

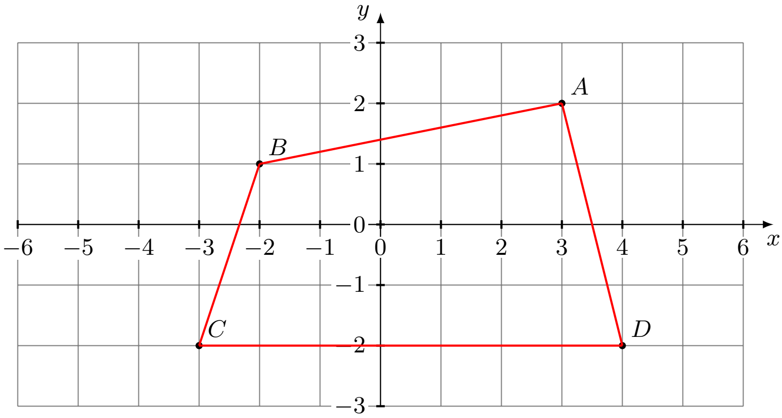
- Number-Systems (hex,oct,binary)
- Algebra (logarithms & functions)
- Trigonometry (sin, cos, tan & triangle-equations)
- Coordinate Systems
- Vectors & Matrices

Coordinate Systems - Points



Points consist out of x and y value. Coordinates of P are $(3, 2)$.

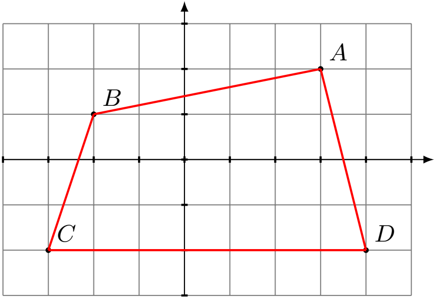
Coordinate Systems - Polygons



We define the Polygon $P = (A, B, C, D)$ [2].

[2] The order is important (in this case the order is counterclockwise)

Coordinate Systems - Area of a Shape

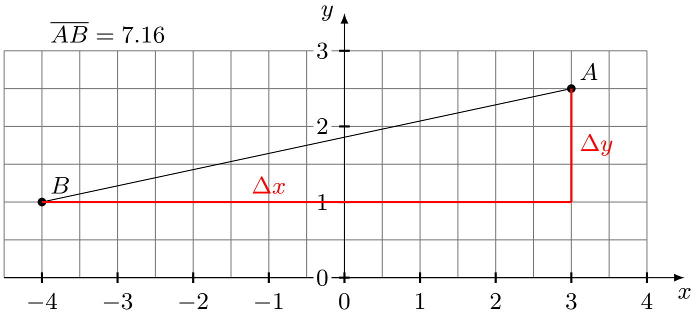


x	y
$x_A = 3$	$y_A = 2$
$x_B = -2$	$y_B = 1$
$x_C = -3$	$y_C = -2$
$x_D = 4$	$y_D = -2$

Area of a 2D Polygon

$area =$
$\frac{1}{2}((x_A y_B - x_B y_A) + (x_B y_C - x_C y_B) + (x_C y_D - x_D y_C) + (x_D y_A - x_A y_D))$

Coordinate Systems - Pythagoras



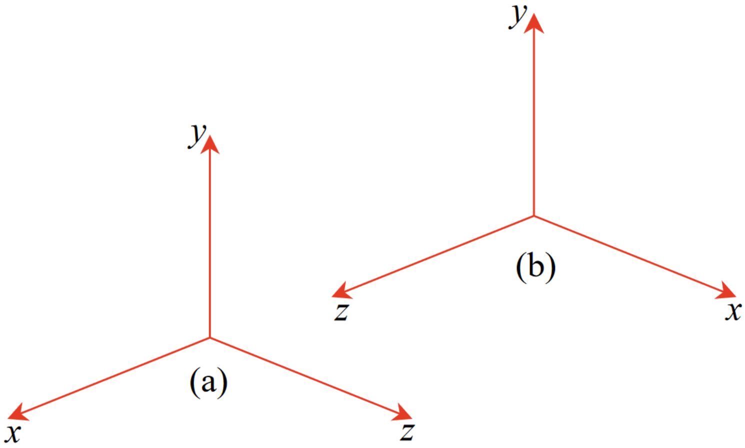
Theorem of Pythagoras

$$distance = |\overline{AB}| = \sqrt{(\Delta x)^2 + (\Delta y)^2} = |\overline{BA}|$$

Distance $\overline{AB} = \sqrt{(3-(-4))^2 + (2.5-1)^2} = \sqrt{7^2 + 1.5^2} \approx 7.16$

3D Coordinates Systems

Extends the 2D coordinate system with a z dimension



(a) := left-handed axial system

(b) := right-handed axial system

Pythagoras in 3D

Theorem of Pythagoras - 2D

$$distance = |\overline{AB}| = \sqrt{(\Delta x)^2 + (\Delta y)^2} = |\overline{BA}|$$

Theorem of Pythagoras - 3D

$$distance = |\overline{AB}| = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} = |\overline{BA}|$$

Vectors

Generic Vectors

Ordered collection off values of same type

In computer graphics $float$ and int vectors are primarily used:

Vector 2D $(int2, uint2, float2, glm :: vec2)$

Two values commonly named (x, y) or (u, v)

Vector 3D $(int3, uint3, float3, glm :: vec3)$

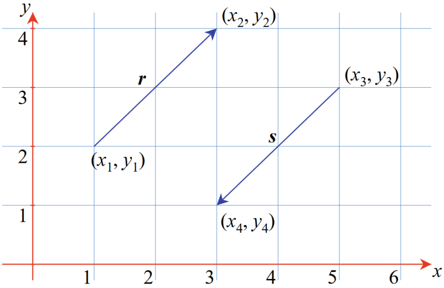
Three values commonly named (x, y, z) or (r, b, g)

Vector 4D $(int4, uint4, float4, glm :: vec4)$

Four values commonly named (x, y, z, w) or (r, g, b, a)

Vector Attributes

Vectors are commonly associated with points in space or directions.



Magnitude of a Vector

$$||r|| = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$

Scaling a vector

Scaling a vector n by a scalar value 2:

$$n = \begin{bmatrix} 2 \\ 6 \\ -2 \end{bmatrix}, \quad \text{then} \quad 2n = \begin{bmatrix} 4 \\ 12 \\ -4 \end{bmatrix}$$

In General:

$$n = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \text{then} \quad \lambda n = \begin{bmatrix} \lambda x \\ \lambda y \\ \lambda z \end{bmatrix}, \quad \text{where} \quad \lambda \in \mathbb{R}.$$

Vector Addition and Subtraction

$$r = \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix}, \quad s = \begin{bmatrix} x_s \\ y_s \\ z_s \end{bmatrix}, \quad \text{then} \quad r \pm s = \begin{bmatrix} x_r \pm x_s \\ y_r \pm y_s \\ z_r \pm z_s \end{bmatrix}$$

Addition is commutative

$$r + s = \begin{bmatrix} x_r + x_s \\ y_r + y_s \\ z_r + z_s \end{bmatrix} = s + r$$

Subtraction is **not** commutative

$$r - s = \begin{bmatrix} x_r - x_s \\ y_r - y_s \\ z_r - z_s \end{bmatrix} \neq s - r$$

Unit Vector

A *unit vector* has a magnitude of 1:

$$i = \textit{unit vector}, \quad \text{when} \quad ||i|| = 1$$

Every[3] vector r can be *normalized* to a *unit vector*:

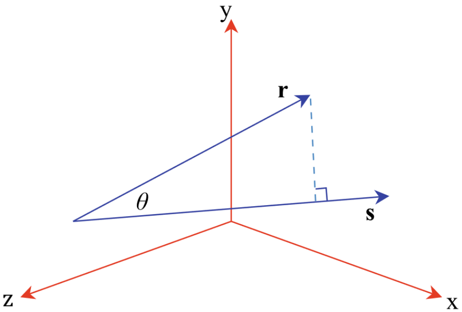
$$\hat{r} = \frac{1}{||r||} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix}$$

[3] Unless the magnitude is equal 0

Scalar[4]/Dot Product

Scalar Product:

$$r \cdot s = ||r|| \cdot ||s|| \cdot \cos\theta$$

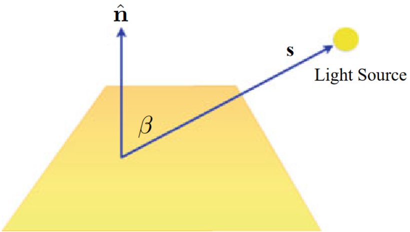


The Scalar Product can be extend to:

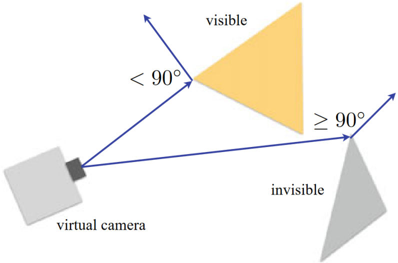
$$r \cdot s = ||r|| \cdot ||s|| \cdot \cos\theta = x_r x_s + y_r y_s + z_r z_s$$

[4] It is called scalar product because it computes a scalar (one dimensional) value.

Dot Product - Examples



Lighting calculations



Backface detection

Cross Product

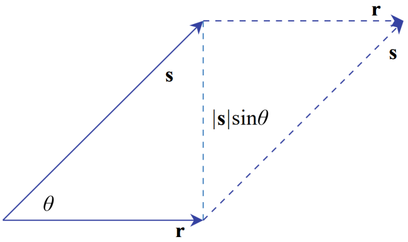
The cross product calculates a new vector t:

$$r \times s = t$$

The Magnitude of t is equal to: $||t|| = ||r|| \cdot ||s|| \cdot \sin\theta$

$$||t|| = ||r|| \cdot ||s|| \cdot \sin\theta$$

which is the area of the parallelogram formed by r and s:



The resulting vector t is perpendicular to the parallelogram[5].
The cross product can be extend to:

Conclusion

Today we (re)learned:

- logarithms
- trigonometry
- coordinate systems
- vectors

Outlook

We still need a bit more math, but we will look at it in a later lecture:

- Matrices
- Affine Transforms
- Interpolation
- Curves
- Barycentric Coordinates

Any Questions?